Airline Sentiment Analysis

Production-Grade Technical Documentation

DistilBERT NLP Model | MLflow-Governed Lifecycle | FastAPI + Streamlit

Final Model Performance (Version 13)

Test Accuracy 82.83% | Macro F1: 0.78 | Weighted F1: 0.83 | Latency: ~0.12s (CUDA)

14,604 Tweets Analyzed | 6 Major Airlines | 3-Class Sentiment Model

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Module 01 - Executive Summary

1.1 The Governing Thought

This project delivers an intelligent, production-ready sentiment analysis system that automatically classifies 14,604 airline customer tweets into Positive, Neutral, and Negative - transforming raw social media noise into structured, actionable business intelligence with 82.83% accuracy and ~0.12s inference latency.

1.2 Key Highlights

- Model Deployed: DistilBERT v13 promoted to Production via MLflow Automated Quality Gate - surpassed all thresholds (Accuracy > 75%, Macro F1 > 0.70).
- Critical Finding: 62.72% of all airline tweets are Negative, with Customer Service Issues driving 31.7% of complaints - a systemic, human-interaction failure, not operational.
- Competitive Insight: Southwest Airlines leads in positive sentiment (23.6%), while United Airlines faces a reputation crisis (68.9% negative). A clear performance gap exists with a replicable blueprint.

1.3 The Ask / Decision

Decision Required	Detail	Priority
Deploy to Production	Approve FastAPI + Streamlit stack for live tweet monitoring	High
Scale Monitoring Team	Staff social media teams 6 AM - 12 PM (peak hours)	High
Budget for Phase 2	Fund expanded training data & multilingual support	Medium

Module 02 - Context & Problem Definition (SCQA)

2.1 Situation

Airlines collectively receive thousands of social media mentions daily. This dataset captures 14,604 tweets targeting 6 major US carriers (United, US Airways, American, Southwest, Delta, Virgin America) - a representative cross-section of real-time public customer sentiment.

Airline	Total Tweets	Negative	Positive
United	3,822	2,633 (68.9%)	492
US Airways	2,913	2,263 (77.7%)	--
American	2,723	1,941 (71.3%)	--
Southwest	2,420	--	570 (23.6%)
Delta	2,222	--	544
Virgin America	504	--	--

2.2 Complication

With negative sentiment at 62.72% and a 4:1 negative-to-positive ratio, airline teams cannot manually process this volume. Critical complaints go unanswered, viral negative content spreads unchecked, and management lacks data-driven insight to prioritize service improvements. Customer service issues - not operational failures - represent the #1 complaint (31.7%), meaning the problem is systemic and human.

2.3 Question

How can we automatically classify customer sentiment at scale - in real-time - with sufficient accuracy to drive operational decisions, escalate urgent cases, and benchmark airline performance?

2.4 Answer

Solution: A production-grade DistilBERT NLP model (Version 13) fine-tuned on airline tweets, served via FastAPI REST API and Streamlit dashboard, governed by an MLflow lifecycle with automated quality gates.
Outcome: Real-time 3-class sentiment classification (Positive / Neutral / Negative) at 82.83% accuracy with ~0.12s latency per tweet - fast enough for live monitoring pipelines.

Module 03 - Project Objective (SMART Goals)

3.1 Primary Objective

Design, train, and deploy a DistilBERT-based sentiment analysis model that classifies airline customer tweets into 3 categories (Positive, Neutral, Negative) with a Test Accuracy >= 75% and Macro F1 >= 0.70, serving predictions via REST API in <= 0.15 seconds, governed by an automated MLflow production lifecycle - delivered within a 4-week development sprint.

3.2 SMART Goal Breakdown

Dimension	Goal	Achieved?
Specific	3-class sentiment classifier for airline tweets using DistilBERT	Yes
Measurable	Test Accuracy >= 75% | Macro F1 >= 0.70	82.83% / 0.78
Achievable	Fine-tune pre-trained DistilBERT - no training from scratch	Yes
Relevant	Transforms raw tweets into business intelligence for airline ops	Yes
Time-bound	Production deployment within project sprint	Version 13 promoted

3.3 Business Goals

Understand Customer Voice: Extract insights from 14,604 tweets to quantify satisfaction levels and pain points across 6 airlines.
Real-time Monitoring: Enable airlines to track brand reputation and customer sentiment on social media as it happens.
Data-Driven Decisions: Provide actionable intelligence for management to improve service quality and operational efficiency.
Competitive Analysis: Benchmark sentiment performance across airlines to identify industry leaders vs. laggards.

3.4 Technical Goals

Automated Classification: Deep learning model for sentiment prediction without manual intervention.
Scalable Architecture: System capable of processing large social media volumes efficiently.
High Accuracy: Reliable predictions that business decisions can depend on with confidence.

Module 04 - Methodology & Architecture

4.1 Tech Stack

Layer	Technology	Purpose
Model	DistilBERT (distilbert-base-uncased)	Core NLP inference engine
API	FastAPI Framework	High-performance REST API
Interface	Streamlit	Interactive monitoring dashboard
Lifecycle	MLflow	Experiment tracking & model versioning
Training	PyTorch + Transformers	Fine-tuning & optimization
Augmentation	NLTK Synonym Replacement	Class imbalance handling

4.2 Why DistilBERT?

Speed: DistilBERT is 40% smaller and 60% faster than BERT-base, achieving ~0.12s inference latency - essential for real-time tweet monitoring pipelines.

Accuracy: Retains 97% of BERT's language understanding capability, making it ideal for nuanced short-text classification like tweets.

Practicality: Unlike GPT-class models, DistilBERT fine-tunes efficiently on domain-specific data (airline vocabulary) without requiring massive compute budgets.

4.3 Version 13 Key Improvements

1

Custom Dataset & Lazy Loading

Instead of tokenizing the entire dataset at once, the model now tokenizes only the required batch on-the-fly during training. This significantly reduces memory overhead and speeds up the data loading pipeline, making it scalable to larger datasets without RAM bottlenecks.

2

Linear Warmup Scheduler

Integrated get_linear_schedule_with_warmup to gradually increase the learning rate during the warmup phase before decaying it linearly. This stabilizes early training steps and prevents large, destabilizing gradient updates at the start of fine-tuning.

```
scheduler = get_linear_schedule_with_warmup(optimizer, num_warmup_steps=...,
num_training_steps=...)
```

3

Gradient Clipping

Applied torch.nn.utils.clip_grad_norm_ to cap gradient magnitudes during backpropagation. This prevents the exploding gradients problem, ensuring stable and consistent weight updates throughout training.

```
torch.nn.utils.clip_grad_norm_(model.parameters(), max_norm=1.0)
```

4

Validation Set Split

The dataset was formally split into three non-overlapping partitions: Train / Validation / Test. This eliminated data leakage between training and evaluation, directly contributing to the improvement in model stability and the increase in Test Accuracy.

5

Class Weights — Loss Adjustment

Class weights (1 : 2 : 1.5) were passed to CrossEntropyLoss to penalize misclassifications of minority classes (Neutral and Positive). This counteracts the majority-class bias caused by the heavily imbalanced dataset (62.72% Negative), improving F1-Scores for underrepresented classes.

```
criterion = CrossEntropyLoss(weight=torch.tensor([1.0, 2.0, 1.5]))
```

Note: Hybrid Optimization combined Synonym-based Data Augmentation with Cost-Sensitive Learning. Class Weights (1:2:1.5) in the Loss Function penalized minority class misclassifications, eliminating majority class bias. The Data Leakage Fix (Improvement #4) directly contributed to the notable improvement in model stability reflected in the increased Test Accuracy.

4.4 Model Specifications

Parameter	Value
Model Architecture	DistilBERT (distilbert-base-uncased)
Batch Size	16
Training Epochs	3
Max Sequence Length	512 tokens
Test Accuracy	82.83%
Macro F1-Score	0.78
Weighted F1-Score	0.83
Inference Latency	~0.12s (CUDA)

4.5 MLOps Pipeline

Component	Description
MLflow Tracking	Experiment tracking and metrics logging
MLflow Packaging	Standardized model artifacts
Model Signature	Input/output schema validation
MLflow Registry	Centralized model versioning
Staging Transition	Pre-production testing environment
Quality Gate	Automated validation before promotion
Production Transition	Controlled production release
Conda.yaml	Dependency & environment management

Module 05 - Analysis & Insights

5.1 Model Performance Metrics

82.83%	0.78	0.83	~0.12s
Test Accuracy	Macro F1-Score	Weighted F1-Score	Inference Latency

Metric	Score	Status
Test Accuracy	82.83%	Passed Gate (> 75%)
Macro F1-Score	0.78	Passed Gate (> 0.70)
Weighted F1-Score	0.83	Production Ready
Inference Latency	~0.12s	Optimized (CUDA)

5.2 Class-wise Performance

Sentiment Class	Precision	Recall	F1-Score	Support	Confidence
Negative	0.89	0.90	0.90	1,377	93.3%
Neutral	0.68	0.65	0.66	465	82.3%
Positive	0.77	0.78	0.78	354	87.2%
Macro Avg	0.78	0.78	0.78	--	--
Weighted Avg	0.83	0.83	0.83	--	--

Synthesis: Negative tweets are easiest to classify (F1: 0.90) - customers use explicit, unambiguous language when complaining. Neutral tweets improved to F1: 0.66 with Recall of 0.65. Positive results reached F1: 0.78 due to hybrid optimization and data leakage fix in v13.

5.3 Overall Sentiment Distribution

Sentiment	Tweet Count	Percentage
Negative	9,159	62.72%
Neutral	3,091	21.17%
Positive	2,354	16.12%

The overwhelming prevalence of negative tweets (nearly two-thirds of all data) indicates significant systemic customer dissatisfaction. The 4:1 negative-to-positive ratio is a critical business risk - dissatisfied customers are far more vocal than satisfied ones, and negative content shows higher retweet rates (0.093 vs 0.070), amplifying reputational damage.

5.4 Airline-Specific Findings

Airline	Total Tweets	Negative	Positive	Top Complaint
United	3,822	2,633 (68.9%)	492	Customer Service

Airline	Total Tweets	Negative	Positive	Top Complaint
US Airways	2,913	2,263 (77.7%)	--	Customer Service
American	2,723	1,941 (71.3%)	--	Customer Service
Southwest	2,420	--	570 (23.6%)	Customer Service
Delta	2,222	--	544	Late Flight
Virgin America	504	--	--	Customer Service

5.5 Complaint Analysis

Complaint Category	Count	% of Negative
Customer Service Issue	2,904	31.7%
Late Flight	1,660	18.1%
Can't Tell	1,190	13.0%
Cancelled Flight	843	9.2%
Lost Luggage	721	7.9%

Synthesis: Customer service failures - not operational issues - are the #1 driver of negative sentiment across 5 of 6 airlines. This points to systemic gaps in frontline training, escalation protocols, and staff empowerment, not just flight operations.

5.6 Tweet Behavioral Patterns

Metric	Negative	Neutral	Positive
Avg. Tweet Length (words)	11.05	8.45	8.52
Retweet Rate	0.093	0.061	0.070
Model Confidence	93.3%	82.3%	87.2%

Negative tweets are longer and more detailed, indicating customers invest significant effort in articulating complaints. Higher retweet rates confirm negative content spreads faster - a viral risk requiring rapid response protocols.

5.7 Temporal & Geographic Patterns

Peak Hours: Tweet activity peaks 6 AM - 9 AM, rising from 131 tweets at midnight to 999 at 9 AM. Morning commutes and early flight disruptions drive complaint spikes.

Top Location: Boston, MA accounts for 33.4% of all tweets (4,880 tweets) - suggesting either data collection bias or Boston as a high-activity hub market.

Power Users: Most active accounts (JetBlueNews: 63, kbosspotter: 32) appear to be aviation enthusiasts or industry monitors, not typical passengers.

Module 06 - Next Steps & Recommendations

6.1 Model Improvement Roadmap

Improvement	Impact	Effort
Expand training data with recent tweets	Higher accuracy on evolving language	Medium
Add multilingual support (Spanish, French)	Broader market coverage	High
Fine-tune on airline-specific vocabulary	Better domain performance	Low
Experiment with RoBERTa / DeBERTa	Potential +2-3% accuracy gain	Medium
Build ensemble model (BERT + LSTM)	Improved Neutral class F1	High

6.2 Deployment Plan

Phase 1 (0-1 month): Productionize FastAPI endpoint + Streamlit dashboard. Configure auto-alerting for high-confidence Negative tweets targeting United and US Airways accounts.

Phase 2 (1-3 months): Integrate with airline CRM systems. Route high-severity complaints directly to service recovery teams. Establish real-time sentiment KPI dashboards for management.

Phase 3 (3-6 months): Expand to Instagram and Facebook data sources. Deploy multilingual model. Build predictive analytics for proactive issue resolution before complaints escalate.

6.3 Strategic Action Plan

Immediate Actions (0-3 months)

Launch 24/7 social media response teams with clear escalation protocols.

Implement staff customer service training programs (empathy + conflict resolution).

Establish real-time sentiment monitoring dashboards powered by this model.

Deploy rapid-response protocol: respond to negative tweets within 1 hour.

Short-term Initiatives (3-6 months)

Roll out service recovery policies with clear compensation guidelines.

Implement baggage tracking technology to reduce lost luggage incidents.

Conduct customer satisfaction surveys to validate social media findings.

Benchmark United Airlines against Southwest's best practices - the gap is measurable.

Long-term Transformation (6-12 months)

Cultural transformation programs focused on customer-centricity across all airlines.

Operational excellence initiatives targeting on-time performance (especially Delta).

Technology investments in predictive analytics for proactive issue resolution.

Develop industry-leading customer service standards with Southwest as the benchmark.

6.4 Resources Required

Resource	Detail	Timeline
Engineering Team	2 ML engineers for model maintenance + API scaling	Ongoing
Social Media Team	4-6 agents staffed 6 AM - 12 PM (peak hours)	Month 1
Infrastructure	GPU server for CUDA inference + MLflow server	Month 1
Training Budget	Data labeling for 5,000+ new tweets per quarter	Quarterly
Analytics Tool	BI dashboard integration (Tableau / Power BI)	Month 2

Airlines that invest in frontline staff empowerment, proactive communication, and genuine service recovery will differentiate themselves in a market where customer expectations are high but satisfaction is consistently low. Southwest Airlines' relatively strong performance (23.6% positive) provides a replicable blueprint - and this model gives every airline the data infrastructure to follow it.

6.5 Key Metrics Summary

Metric	Value
Test Accuracy	82.83%
Macro F1-Score	0.78
Weighted F1-Score	0.83
Inference Latency	~0.12s (CUDA)
Total Tweets Analyzed	14,604
Negative Sentiment	62.72% (9,159 tweets)
Top Complaint	Customer Service (31.7%)
Most Negative Airline	US Airways (77.7% negative)
Most Positive Airline	Southwest (23.6% positive)
Peak Activity Time	6 AM - 9 AM
Report Version	Version 13 - December 2025